

Directions. Q. Nos. (115-117) Let $\sin(A+B)=1$ and $\sin(A-B)=\frac{1}{2}$, where $A, B \in \left[0, \frac{\pi}{2}\right]$.

- 115.** What is the value of A ?
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{8}$
- 116.** What is the value of $\tan(A+2B)\tan(2A+B)$?
 (a) -1 (b) 0 (c) 1 (d) 2
- 117.** What is the value of $\sin^2 A - \sin^2 B$?
 (a) 0 (b) $1/2$ (c) 1 (d) 2

Directions. Q. Nos. (118-120) ABC is an obtuse triangle and $\tan A, \tan B$ are the roots of the equation $3x^2 - 2\sqrt{3}x + 1 = 0$.

- 118.** The measure of angle C is
 (a) $\pi/3$ (b) $\pi/2$ (c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$
- 119.** Find the value of $\sin C + \cos C$.
 (a) $\frac{2}{\sqrt{3}+1}$ (b) $\frac{\sqrt{3}+1}{2}$ (c) $\frac{\sqrt{3}-1}{2}$ (d) $\sqrt{3} + \frac{1}{2}$
- 120.** If $2\sin B = \sqrt{\frac{1+\tan^2 C}{\operatorname{cosec}^2 C}}$, then the value of B is
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{6}$

PREVIOUS YEARS' QUESTIONS

- 121.** The expression $\sin^2 x + \cos^2 x - 1 = 0$ is satisfied by how many values of x ? **2012 I**
 (a) Only one value of x (b) Two values of x
 (c) Infinite values of x (d) No value of x
- 122.** If $3\sin x + 5\cos x = 5$, then what is the value of $(3\cos x - 5\sin x)$? **2012 I**
 (a) 0 (b) 2 (c) 3 (d) 5
- 123.** If $p = a\sin x + b\cos x$ and $q = a\cos x - b\sin x$, then what is the value of $p^2 + q^2$? **2012 I**
 (a) $a+b$ (b) ab (c) $a^2 + b^2$ (d) $a^2 - b^2$
- 124.** If α and β are complementary angles, then what is $\sqrt{\operatorname{cosec} \alpha \cdot \operatorname{cosec} \beta} \left(\frac{\sin \alpha}{\sin \beta} + \frac{\cos \alpha}{\cos \beta} \right)^{-\frac{1}{2}}$ equal to? **2012 I**
 (a) 0 (b) 1
 (c) 2 (d) None of these
- 125.** Consider the following statements
 I. $\frac{\cot 30^\circ + 1}{\cot 30^\circ - 1} = 2(\cos 30^\circ + 1)$
 II. $2\sin 45^\circ \cos 45^\circ - \tan 45^\circ \cot 45^\circ = 0$

Which of the above statement(s) is/are correct?

- (a) Only I (b) Only II **2012 I**
 (c) Both I and II (d) Neither I nor II

- 126.** Consider the following statements

- I. $\sin^2 1^\circ + \cos^2 1^\circ = 1$
 II. $\sec^2 33^\circ - \cot^2 57^\circ = \operatorname{cosec}^2 37^\circ - \tan^2 53^\circ$

Which of the above statement(s) given above is/are correct? **2012 I**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 127.** Consider the following statements:

- I. There is only one value of x in the first quadrant that satisfies $\sin x + \cos x = 2$.
 II. There is only one value of x in the first quadrant that satisfies $\sin x - \cos x = 0$.

Which of the above statement(s) given above is/are correct? **2012 I**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 128.** If $\sin \theta \cos \theta = \sqrt{3}/4$, then $\sin^4 \theta + \cos^4 \theta$ is equal to **2012 II**

- (a) $7/8$ (b) $5/8$ (c) $3/8$ (d) $1/8$

- 129.** If the angle θ is in the first quadrant and $\tan \theta = 3$, then what is the value of $(\sin \theta + \cos \theta)$? **2012 II**

- (a) $\frac{1}{\sqrt{10}}$ (b) $\frac{2}{\sqrt{10}}$ (c) $\frac{3}{\sqrt{10}}$ (d) $\frac{4}{\sqrt{10}}$

- 130.** If $0^\circ < \theta < 90^\circ$, then all the trigonometric ratios can be obtained when **2012 II**

- (a) only $\sin \theta$ is given
 (b) only $\cos \theta$ is given
 (c) only $\tan \theta$ is given
 (d) any one of the six ratios is given

- 131.** What is the value of $\sin A \cos A \tan A + \cos A \sin A \cot A$? **2012 II**

- (a) $\sin^2 A + \cos^2 A$ (b) $\sin^2 A + \tan^2 A$
 (c) $\sin^2 A + \cot^2 A$ (d) $\operatorname{cosec}^2 A + \cot^2 A$

- 132.** What is the value of $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta}$? **2012 II**

- (a) $2 \operatorname{cosec} \theta$ (b) $2 \sec \theta$
 (c) $\sec \theta$ (d) $\operatorname{cosec} \theta$

- 133.** What is the value of $\sec^2 D - \tan^2 D$? **2013 I**

- (a) $1/2$ (b) $2/3$
 (c) 1 (d) None of these

- 134.** If $\cos A + \cos^2 A = 1$, then what is the value of $2(\sin^2 A + \sin^4 A)$? **2013 I**

- (a) 4 (b) 2 (c) 1 (d) $1/2$

- 135.** $(1 - \tan A)^2 + (1 + \tan A)^2 + (1 - \cot A)^2 + (1 + \cot A)^2$ is equal to ☑ 2013 I

(a) $\sin^2 A \cos^2 A$ (b) $\sec^2 A \operatorname{cosec}^2 A$
(c) $2 \sec^2 A \operatorname{cosec}^2 A$ (d) None of these

- 136.** If $\alpha^2 = \frac{1 + 2 \sin \theta \cos \theta}{1 - 2 \sin \theta \cos \theta}$, then what is the value of $\frac{\alpha + 1}{\alpha - 1}$? ☑ 2013 I

(a) $\sec \theta$ (b) 1 (c) 0 (d) $\tan \theta$

- 137.** If $\sin \theta = \frac{x^2 - y^2}{x^2 + y^2}$, then which one of the following is correct? ☑ 2013 I

(a) $\cos \theta = \frac{2xy}{x^2 - y^2}$ (b) $\cos \theta = \frac{2xy}{x^2 + y^2}$
(c) $\cos \theta = \frac{x - y}{x^2 + y^2}$ (d) $\cos \theta = \frac{xy(x - y)}{x^2 + y^2}$

- 138.** Consider the following statements for $0 \leq \theta \leq 90^\circ$:
- I. The value of $\sin \theta + \cos \theta$ is always greater than 1.
II. The value of $\tan \theta + \cot \theta$ is always greater than 1.
- Which of the statement(s) given above is/are correct?
- (a) Only I (b) Only II ☑ 2013 I
(c) Both I and II (d) Neither I nor II

Directions (Q. Nos. 139-141) Read the following information carefully to answer the questions that follow.

The angles A, B, C and D of a quadrilateral ABCD are in the ratio 1 : 2 : 4 : 5.

- 139.** What is the value of $\cos(A + B)$? ☑ 2013 I
- (a) 0 (b) $1/2$
(c) 1 (d) None of these

- 140.** What is the value of $\operatorname{cosec}(C - D + B)$? ☑ 2013 I
- (a) 1 (b) 2 (c) 3 (d) 4

- 141.** Consider the following statements:
- I. ABCD is a cyclic quadrilateral.
II. $\sin(B - A) = \cos(D - C)$
- Which of the statement(s) given above is/are correct? ☑ 2013 I
- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 142.** If $\sin \theta + \cos \theta = \sqrt{3}$, then what is $\tan \theta + \cot \theta$ equal to? ☑ 2013 II
- (a) 1 (b) $\sqrt{2}$ (c) 2 (d) $\sqrt{3}$

- 143.** What is $\operatorname{cosec}(75^\circ + \theta) - \sec(15^\circ - \theta)$ equal to? ☑ 2013 II

(a) 0 (b) 1 (c) $2 \sin \theta$ (d) $2 \cos \theta$

- 144.** If $5 \sin \theta + 12 \cos \theta = 13$, then what is $5 \cos \theta - 12 \sin \theta$ equal to? ☑ 2013 II
- (a) -2 (b) -1 (c) 0 (d) 1

- 145.** If $\sin \theta - \cos \theta = 0$, then what is $\sin^4 \theta + \cos^4 \theta$ equal to? ☑ 2013 II
- (a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

- 146.** What is $\frac{\cos^2(45^\circ + \theta) + \cos^2(45^\circ - \theta)}{\tan(60^\circ + \theta) \tan(30^\circ - \theta)}$ equal to? ☑ 2013 II
- (a) -1 (b) 0 (c) 1 (d) 2

- 147.** If $\tan \theta + \sec \theta = m$, then what is $\sec \theta$ equal to? ☑ 2013 II
- (a) $\frac{m^2 - 1}{2m}$ (b) $\frac{m^2 + 1}{2m}$ (c) $\frac{m + 1}{m}$ (d) $\frac{m^2 + 1}{m}$

- 148.** If α, β and γ are acute angles such that $\sin \alpha = \frac{\sqrt{3}}{2}$, $\cos \beta = \frac{\sqrt{3}}{2}$ and $\tan \gamma = 1$, then what is $\alpha + \beta + \gamma$ equal to? ☑ 2013 II
- (a) 105° (b) 120°
(c) 135° (d) 150°

- 149.** What is $\frac{(1 + \sec \theta - \tan \theta) \cos \theta}{(1 + \sec \theta + \tan \theta)(1 - \sin \theta)}$ equal to? ☑ 2013 II
- (a) 1 (b) 2 (c) $\tan \theta$ (d) $\cot \theta$

- 150.** If $\triangle ABC$ is right angled at C, then what is $\cos(A + B) + \sin(A + B)$ equal to? ☑ 2013 II
- (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) 2

- 151.** Consider the following statements:
- I. $\tan \theta$ increases faster than $\sin \theta$ as θ increases.
II. The value of $\sin \theta + \cos \theta$ is always greater than 1.
- Which of the statement(s) given above is/are correct? ☑ 2013 II
- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 152.** The value of $\cos 25^\circ - \sin 25^\circ$ is ☑ 2014 I
- (a) positive but less than 1
(b) positive but greater than 1
(c) negative
(d) 0

- 153.** In a right angled $\triangle ABC$, right angle at B, if $\cos A = \frac{4}{5}$, then what is $\sin C$ is equal to? ☑ 2014 I
- (a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{3}{4}$ (d) $\frac{2}{5}$

- 154.** If α and β are complementary angles, then what is $\sqrt{\cos\alpha \operatorname{cosec}\beta - \cos\alpha \sin\beta}$ equal to? **2014 I**
 (a) $\sec\beta$ (b) $\cos\alpha$ (c) $\sin\alpha$ (d) $-\tan\beta$
- 155.** If $\sec\theta + \tan\theta = 2$, then what is the value of $\sec\theta$? **2014 I**
 (a) $\frac{3}{2}$ (b) $\sqrt{2}$ (c) $\frac{5}{2}$ (d) $\frac{5}{4}$
- 156.** What is $\operatorname{cosec}(75^\circ + \theta) - \sec(15^\circ - \theta) - \tan(55^\circ + \theta) + \cot(35^\circ - \theta)$ equal to? **2014 I**
 (a) -1 (b) 0 (c) 1 (d) $\frac{3}{2}$
- 157.** What is $\sin 25^\circ \sin 35^\circ \sec 65^\circ \sec 55^\circ$ equal to? **2014 I**
 (a) -1 (b) 0 (c) $\frac{1}{2}$ (d) 1
- 158.** If $2 \cot\theta = 3$, then what is $\frac{2 \cos\theta - \sin\theta}{2 \cos\theta + \sin\theta}$ equal to? **2014 I**
 (a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$
- 159.** If $\sin\theta \cos\theta = \frac{1}{2}$, then what is $\sin^6\theta + \cos^6\theta$ equal to? **2014 I**
 (a) 1 (b) 2 (c) 3 (d) $\frac{1}{4}$
- 160.** If $\cos x + \sec x = 2$, then what $\cos^n x + \sec^n x$ equal to, where n is a positive integer? **2014 I**
 (a) 2 (b) 2^{n-2} (c) 2^{n-1} (d) 2^n
- 161.** If $\sin\theta + 2 \cos\theta = 1$, where $0 < \theta < \pi/2$, then what is $2 \sin\theta - \cos\theta$ equal to? **2014 I**
 (a) -1 (b) $1/2$ (c) 2 (d) 1
- 162.** If $\tan 8\theta = \cot 2\theta$, where $0 < 8\theta < \frac{\pi}{2}$, then what is the value of $\tan 5\theta$? **2014 I**
 (a) $\frac{1}{\sqrt{3}}$ (b) 1 (c) $\sqrt{3}$ (d) 0
- 163.** If $\sin(A+B) = 1$, where $0 < B < 45^\circ$, then what is $\cos(A-B)$ equal to? **2014 I**
 (a) $\sin 2B$ (b) $\sin B$ (c) $\cos 2B$ (d) $\cos B$
- 164.** At what point of time after 3 O'clock, hour hand and the minute hand of a clock occur at right angles for the first time? **2014 I**
 (a) 9 O'clock (b) 4 h $37\frac{1}{6}$ min
 (c) 3 h $30\frac{8}{11}$ min (d) 3 h $32\frac{8}{11}$ min
- 165.** If $\tan\theta + \cot\theta = 2$, then what is $\sin\theta + \cos\theta$ equal to? **2014 II**
 (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{2}$ (d) 1
- 166.** What is $\frac{\sec x}{\cot x + \tan x}$ equal to? **2014 II**
 (a) $\sin x$ (b) $\cos x$ (c) $\tan x$ (d) $\cot x$
- 167.** What is $(1 + \cot x - \operatorname{cosec} x)(1 + \tan x + \sec x)$ equal to? **2014 II**
 (a) 1 (b) 2 (c) $\sin x$ (d) $\cos x$
- 168.** What is $(\operatorname{cosec} x - \sin x)(\sec x - \cos x)(\tan x + \cot x)$ equal to? **2014 II**
 (a) $\sin x + \cos x$ (b) $\sin x - \cos x$
 (c) 2 (d) 1
- 169.** What is $\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1}$ equal to? **2014 II**
 (a) $\frac{\sin x - 1}{\cos x}$ (b) $\frac{\sin x + 1}{\cos x}$ (c) $\frac{\sin x - 1}{\cos x + 1}$ (d) $\frac{\sin x + 1}{\cos x + 1}$
- 170.** What is $(\sin^2 x - \cos^2 x)(1 - \sin^2 x \cos^2 x)$ equal to? **2014 II**
 (a) $\sin^4 x - \cos^4 x$ (b) $\sin^6 x - \cos^6 x$
 (c) $\cos^8 x - \sin^8 x$ (d) $\sin^8 x - \cos^8 x$
- 171.** What is $(\sin x \cos y + \cos x \sin y)(\sin x \cos y - \cos x \sin y)$ equal to? **2014 II**
 (a) $\cos^2 x - \cos^2 y$ (b) $\cos^2 x - \sin^2 y$
 (c) $\sin^2 x - \cos^2 y$ (d) $\sin^2 x - \sin^2 y$
- 172.** If $A + B + C = 180^\circ$, then $\cot A \cot B + \cot B \cot C + \cot C \cot A$ is equal to **2014 II**
 (a) -1 (b) 2 (c) π (d) 1
- 173.** If $\sin x + \operatorname{cosec} x = 2$, then what is $\sin^9 x + \operatorname{cosec}^9 x$ equal to? **2014 II**
 (a) 2 (b) 18 (c) 512 (d) 1024
- 174.** If $\sin x + \cos x = p$ and $\sin^3 x + \cos^3 x = q$, then what is $p^3 - 3p$ equal to? **2014 II**
 (a) 0 (b) $-2q$ (c) $2q$ (d) $4q$
- 175.** Consider the following statements **2014 II**
 I. $\sin 1^\circ > \sin 1$ II. $\cos 1^\circ < \cos 1$
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 176.** The value of $\operatorname{cosec}^2 67^\circ + \sec^2 57^\circ - \cot^2 33^\circ - \tan^2 23^\circ$ is **2015 I**
 (a) $2\sqrt{2}$ (b) 2 (c) $\sqrt{2}$ (d) 0
- 177.** If $\tan(A+B) = \sqrt{3}$ and $\tan A = 1$, then $\tan(A-B)$ is equal to **2015 I**
 (a) 0 (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{2}$
- 178.** If $\tan A + \cot A = 4$, then $\tan^4 A + \cot^4 A$ is equal to **2015 I**
 (a) 110 (b) 191 (c) 80 (d) 194

- 179.** ABC is a triangle right angled at B and $AB:BC = 3:4$. What is $\sin A + \sin B + \sin C$ equal to? ☑ 2015 I
 (a) 2 (b) $\frac{11}{5}$ (c) $\frac{12}{5}$ (d) 3
- 180.** If $\sin x + \cos x = C$, then $\sin^6 x + \cos^6 x$ is equal to ☑ 2015 I
 (a) $\frac{1+6C^2-3C^4}{16}$ (b) $\frac{1+6C^2-3C^4}{4}$
 (c) $\frac{1+6C^2+3C^4}{16}$ (d) $\frac{1+6C^2+3C^4}{4}$
- 181.** If $\frac{3-\tan^2 A}{1-3\tan^2 A} = k$, where k is a real number, then $\operatorname{cosec} A(3 \sin A - 4 \sin^3 A)$ is equal to ☑ 2015 I
 (a) $\frac{2k}{k-1}$ (b) $\frac{2k}{k-1}$, where $\frac{1}{3} \leq k \leq 3$
 (c) $\frac{2k}{k-1}$, where $k < \frac{1}{3}$ or $k > 3$ (d) $\frac{2k}{k+1}$
- 182.** If $p = \sqrt{\frac{1-\sin x}{1+\sin x}}$, $q = \frac{1-\sin x}{\cos x}$, $r = \frac{\cos x}{1+\sin x}$, then which of the following is/are correct? ☑ 2015 I
 I. $p = q = r$ II. $p^2 = qr$
Select the correct answer using the codes given below
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 183.** Consider the following identity
 I. $\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A} = \sin A + \cos A$
 II. $(1-\sin A - \cos A)^2 = 2(1-\sin A)(1+\cos A)$
Which of the above identity/identities is/are correct?
 (a) Only I (b) Only II ☑ 2015 I
 (c) Both I and II (d) Neither I nor II
- 184.** If a and b are positive, then the relation $\sin \theta = \frac{2a+3b}{3b}$ is ☑ 2015 II
 (a) not possible (b) possible, if $a = b$
 (c) possible, if $a > b$ (d) possible, if $a < b$
- 185.** The minimum value of $\cos^2 x + \cos^2 y - \cos^2 z$ is ☑ 2015 II
 (a) -1 (b) 0
 (c) 2 (d) 2
- 186.** If $\tan \theta + \sec \theta = 2$, then $\tan \theta$ is equal to ☑ 2015 II
 (a) $\frac{3}{4}$ (b) $\frac{5}{4}$ (c) $\frac{3}{2}$ (d) $\frac{5}{2}$
- 187.** $\frac{\cos \theta}{1-\sin \theta}$ is equal to $\left(\text{where, } \theta \neq \frac{\pi}{2} \right)$ ☑ 2015 II
 (a) $\frac{\tan \theta - 1}{\tan \theta + 1}$ (b) $\frac{1+\sin \theta}{\cos \theta}$ (c) $\frac{\tan \theta + 1}{\tan \theta - 1}$ (d) $\frac{1+\cos \theta}{\sin \theta}$
- 188.** If $\tan(x+40)^\circ \tan(x+20)^\circ \tan(3x)^\circ \tan(70-x)^\circ \tan(50-x)^\circ = 1$, then the value of x is equal to ☑ 2015 II
 (a) 30 (b) 20 (c) 15 (d) 10
- 189.** The value of $32 \cot^2 \left(\frac{\pi}{4} \right) - 8 \sec^2 \left(\frac{\pi}{3} \right) + 8 \cos^3 \left(\frac{\pi}{6} \right)$ is equal to ☑ 2015 II
 (a) $\sqrt{3}$ (b) $2\sqrt{3}$ (c) 3 (d) $3\sqrt{3}$
- 190.** If $x = a \cos \theta$ and $y = b \cot \theta$, then $(ax^{-1} - by^{-1})(ax^{-1} + by^{-1})$ is equal to ☑ 2015 II
 (a) 0 (b) 1 (c) $\tan^2 \theta$ (d) $\sin^2 \theta$
- 191.** If θ is an acute angle and $\sin \theta \cos \theta = 2 \cos^3 \theta - 1.5 \cos \theta$, then what is $\sin \theta$ equal to? ☑ 2015 II
 (a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{1-\sqrt{5}}{4}$ (c) $\frac{\sqrt{5}+1}{4}$ (d) $-\frac{\sqrt{5}+1}{4}$
- 192.** A clock is started at noon by 10 minutes past 5, through what angle, the hour hand moves? ☑ 2015 II
 (a) 160° (b) 145° (c) 150° (d) 155°
- 193.** Consider the following statements ☑ 2015 II
 I. $\sin 66^\circ$ is less than $\cos 66^\circ$.
 II. $\sin 26^\circ$ is less than $\cos 26^\circ$.
Which of the above statement(s) is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 194.** Consider the following statements.
 I. $\frac{1+\tan^2 \theta}{1+\cot^2 \theta} = \left(\frac{1-\tan \theta}{1-\cot \theta} \right)^2$ is true for all $0 < \theta < \frac{\pi}{2}, \theta \neq \frac{\pi}{4}$
 II. $\cot \theta = \frac{1}{\tan \theta}$ is true for $\theta = 45^\circ$ only.
Which of the above statements is/are correct? ☑ 2015 II
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 195.** In a $\triangle ABC$, if $A - B = \frac{\pi}{2}$, then $C + 2B$ is equal to ☑ 2016 I
 (a) $\frac{2\pi}{3}$ (b) $\frac{3\pi}{4}$ (c) π (d) $\frac{\pi}{2}$
- 196.** $\left(\frac{\sin 35^\circ}{\cos 55^\circ} \right)^2 - \left(\frac{\cos 55^\circ}{\sin 35^\circ} \right)^2 + 2 \sin 30^\circ$ is equal to ☑ 2016 I
 (a) -1 (b) 0 (c) 1 (d) 2
- 197.** If $\tan \theta + \cot \theta = \frac{4}{\sqrt{3}}$, where $0 < \theta < \frac{\pi}{2}$, then $\sin \theta + \cos \theta$ is equal to ☑ 2016 I
 (a) 1 (b) $\frac{\sqrt{3}-1}{2}$ (c) $\frac{\sqrt{3}+1}{2}$ (d) $\sqrt{2}$

